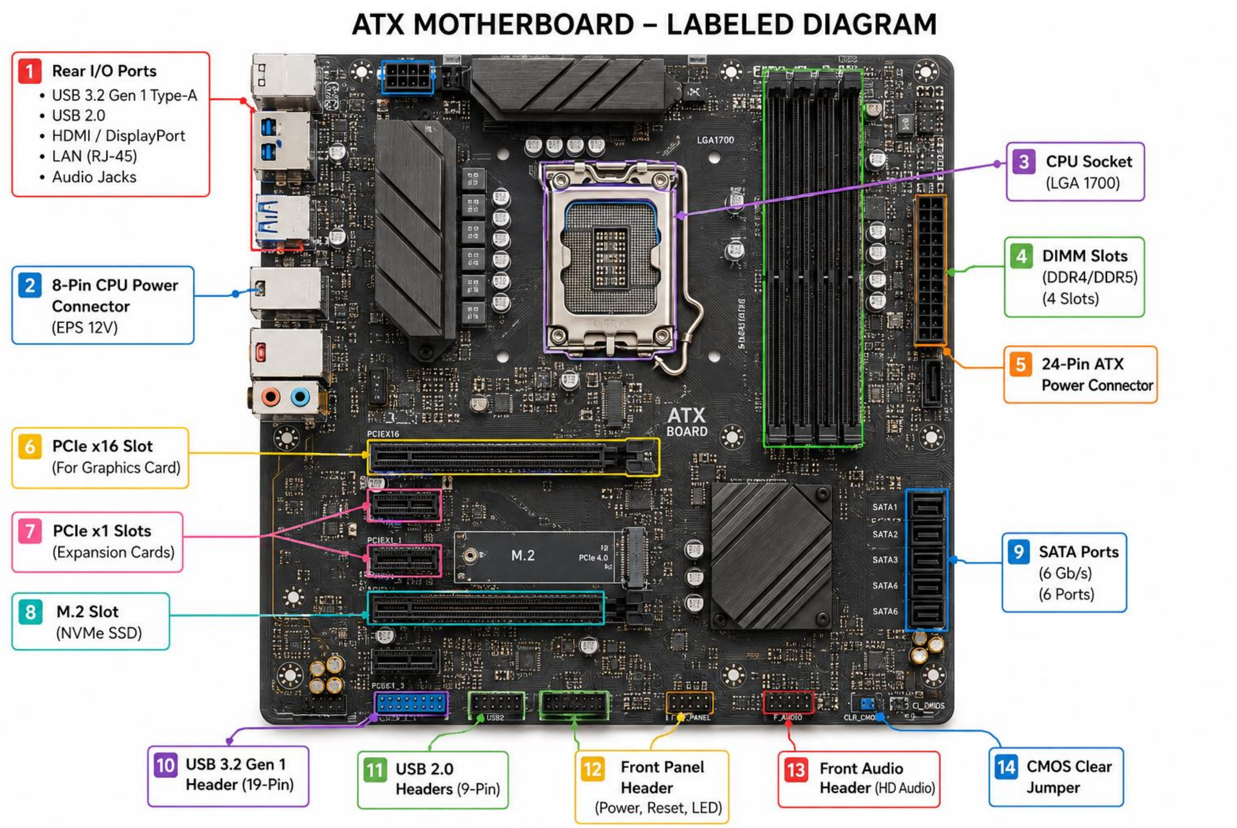
★ **Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).**



2. **Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.**

CPU SOCKET	DESCRIPTION	EX CPU
1 LGA1151 	Intel Socket used with 6th, 7th, 8th, and 9th Gen Processor.	Intel Core i-8700 (8-Gen) 
2 AM4 	AMD Socket used with Ryzen 1000,2000,3000, 4000, and 5000 series Processors. Very popular and long - Latic Socket .	**AMD RYZEN 5 5600X** (Ryzen 5000 series) 
3 LGA1200 	Intel socket for 10th and 11th Gen Processors. Common in Gaming and Productivity PCs.	Intel Core i5 – 10400 (10th Gen) 

3. **Explain in your own words the primary functions of the CPU and the chipset on a motherboard and describe how they work together when you open a music app like Spotify.**

- ➢ The **CPU (Central Processing Unit)** is the **brain of the computer**. It does all the thinking, calculations, and runs programs.

- ➢ It works using the **Fetch–Decode–Execute cycle**, which means it gets instructions, understands them, executes them, and repeats this process very quickly.

- ➢ The **chipset** is like a **traffic controller** on the motherboard. It connects the CPU with the RAM, SSD/HDD, internet, speakers, and other parts. It controls how data moves between all these devices.

❖ Here are the 3 main things a chipset does:

1. Connects the CPU to Everything Else
2. Controls Data Flow
3. Determines Motherboard Features

☺ How they work together when you open Spotify:

- ★ **Step 1:** You click the Spotify icon.

- ★ **Step 2:** The CPU tells the chipset to open Spotify.

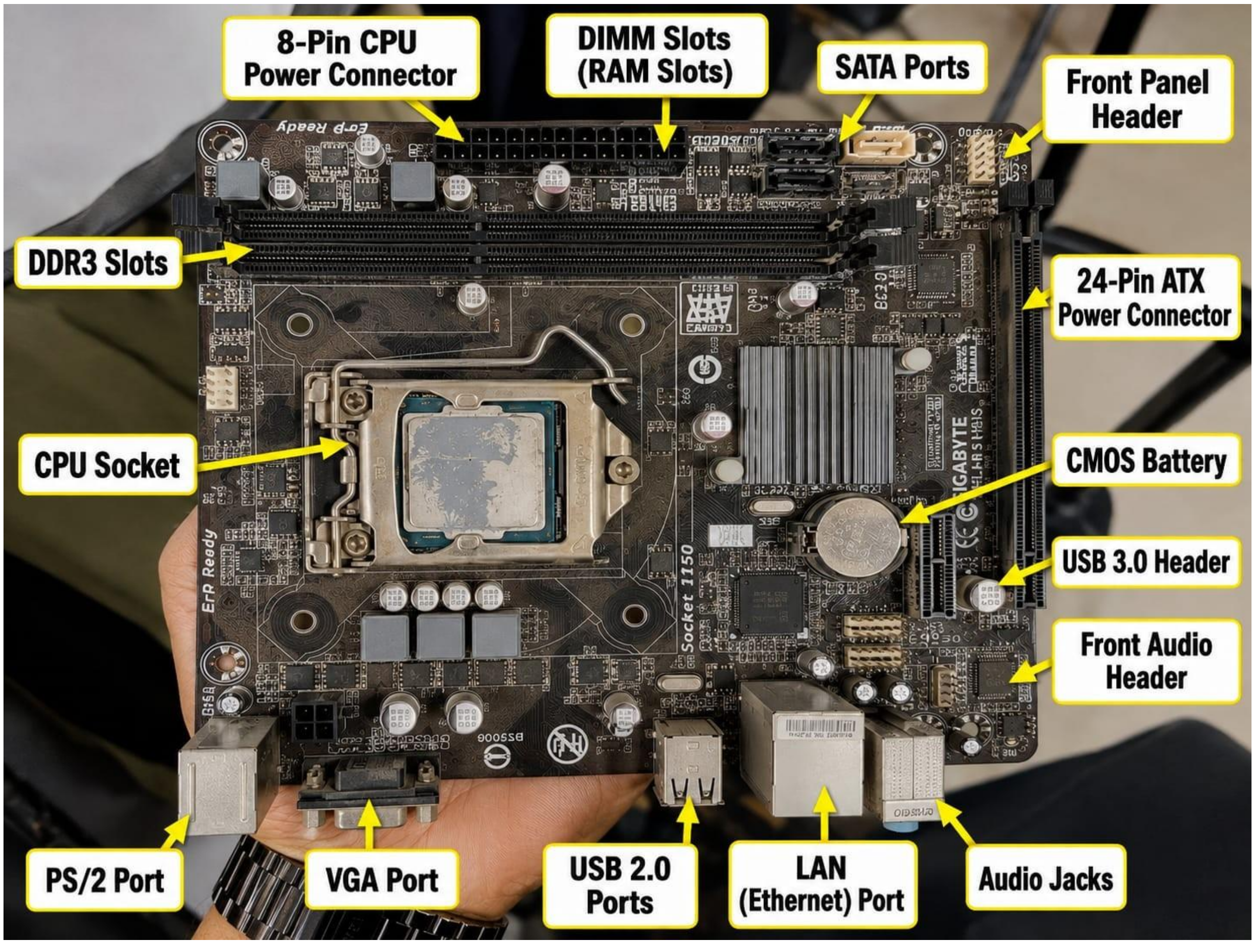
- ★ **Step 3:** The chipset loads the Spotify files from the SSD/HDD into the RAM.

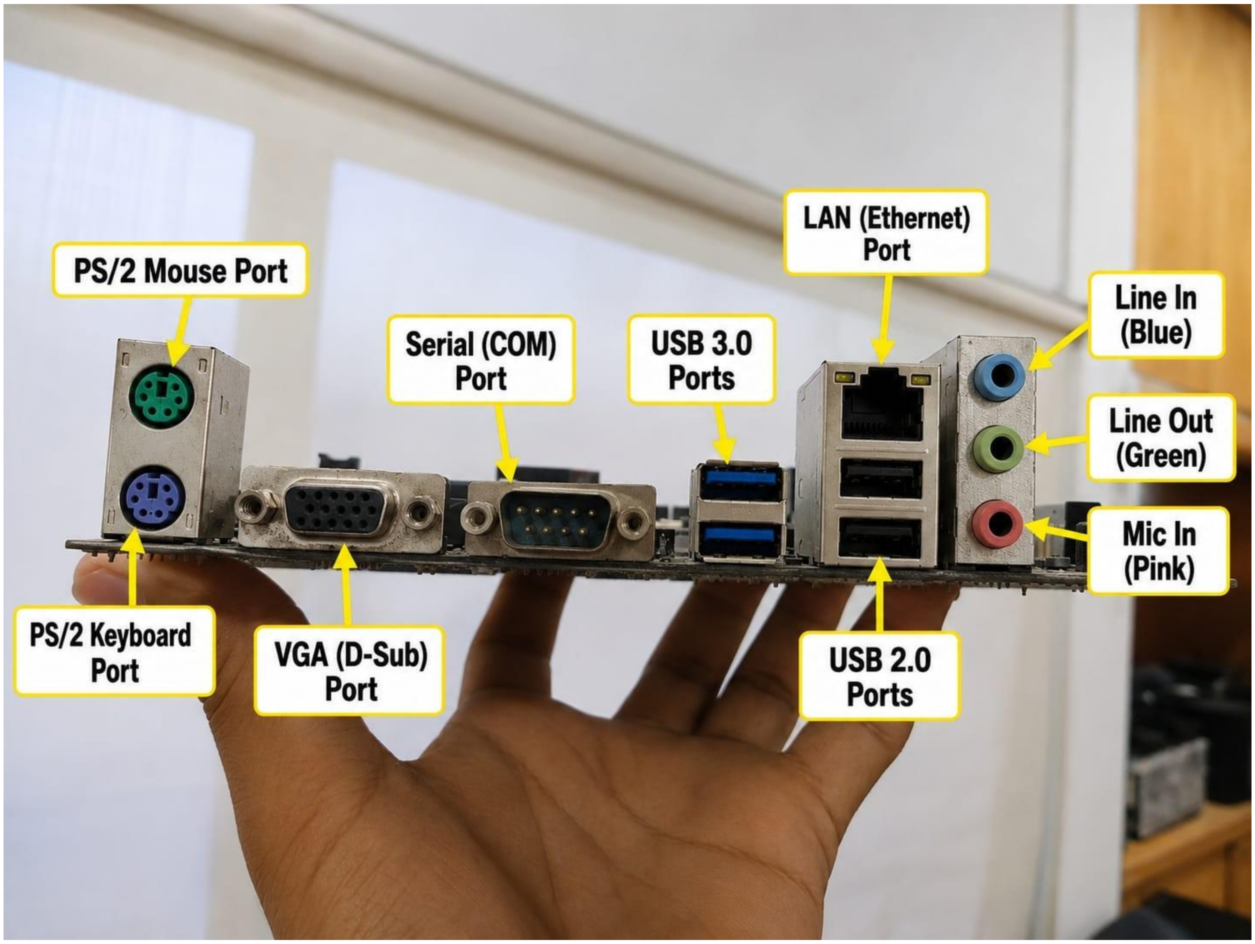
- ★ **Step 4:** The CPU reads the files from the RAM and opens the Spotify app.

★ **Step 5:** When you play a song, the chipset gets the music from the internet.

★ **Step 6:** The CPU processes the music, and the chipset sends the sound to the speakers.

4. **Find a clear photo of a real motherboard (from the internet or your own device) and identify at least 5 different ports or slots by circling them and labeling each one.**




PS/2 Mouse Port
Serial (COM) Port
USB 3.0 Ports
LAN (Ethernet) Port
Line In (Blue)
Line Out (Green)
Mic In (Pink)
PS/2 Keyboard Port
VGA (D-Sub) Port
USB 2.0 Ports